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RESEARCH ARTICLE

Dynamic analysis of cooling tower considering two different column modeling approaches

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Abstract
Nuclear power plants use cooling towers, which are frequently hyperboloid in shape, to disperse excess heat created during power generation. Due to their ever-increasing size, the stability analysis by experimental means and finite element modelling has become more significant. There are different finite element modelling approaches. Often the columns are modelled as simple beam elements to save computational power and analysis duration. In other cases, the columns are modelled with a more realistic approach, thus modelling with the rebars embedded in the concrete with properties like bond and shrinkage. Little attention has been shed on the effect of these two different modelling approaches on the analysis results. In this study, two models of a hyperbolic natural draught cooling tower were made. One model was created with the more realistic Embedded Reinforcement Columns and the other was with Simplified Columns modelled with Timoshenko Beam-type members. The model was validated, and its natural frequency was identified. Subsequently, both models were exposed to seismic and wind excitations, and their responses were compared. The Embedded Reinforcement Column model exhibited greater flexibility and reduced stiffness, resulting in lower natural frequencies. As a consequence, it experienced larger displacements under both seismic and wind loads. While this model demonstrated improved stress dissipation during seismic analysis, it showed less favorable performance under wind loading. The findings of this study are important and relevant given Bangladesh's increased seismic susceptibility, especially in the area of the Madhupur Fault near the Rooppur Nuclear Power Plant. This study can thus help others in the judgement of which modelling approach is suitable for a particular research or analysis.

Keywords Cooling tower · Conventional shell · Abaqus · Modal analysis · Nonlinear time history analysis

1 Introduction

Cooling towers have been a significant part of power stations worldwide for years. Especially, the iconic hyperbolic cooling towers have become ever more popular in the case of dissipating heat from the cooling water in the powerplants. Most of the time, these hyperbolic cooling towers are made of reinforced concrete and supported by columns which can be inclined in a range of angles. Because of the small thickness and large height, proof of their structural integrity is of great importance as any large enough inconvenience may cause a collapse of the tower. Following the 1965 structural failures at Ferrybridge and Ardeer, the need for research on the stability and integrity of these colossal towers has become more significant [1].The evolution of cooling tower shell analysis began with shell bending theory in 1967, leading to the adoption of the Finite Element Method (FEM) in the 1970s for modelling complex structures. In 1983, this study [2] reported on a comprehensive numerical of an RC hyperbolic cooling tower at Port Gibson, Miss. USA, based on the Finite Element Method (FEM). This study [3] introduced us to the early stage of earthquake analysis which concluded that the response spectrum analysis is the best fit. But since then, few works have been done on the seismic analysis of the cooling towers. Due to the ever-increasing size of these towers, response spectrum analysis is not enough these days, which leads us to the work done by [4] where the

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dynamic analysis of the tower was conducted with the linear and non-linear analysis in response with seismic analysis. Although works are scarce regarding seismic analysis of the cooling towers, researches have been done for damage of towers due to dynamic loads in the works of [5–8]. These studies concentrated on the effects of seismic excitation on the columns. A study [9] was carried out with a shaking table test for a proposed natural draft cooling tower, which is 252 m high. Structural dynamic responses corresponding to different levels of seismic actions were measured and analyzed. The results show that columns were the first to be damaged under strong-motion earthquakes. The study conducted by [4] also showed the formation of the plastic hinges at the column-shell junction. Bangladesh is located in a seismically active region, and historical data indicate a significant earthquake is overdue, with the last major one occurring over a century ago. Particularly, the proximity of the Madhupur Fault to the Rooppur Nuclear Power Plant underscores the urgent need for comprehensive seismic analysis of such vital infrastructure.

Elasto-plastic analysis of CFRP strengthened RC slabs reveals improved performance with a CO_2 trade off, while shakedown analysis of long piles under lateral loads enhances design reliability through probabilistic strain energy limits together advancing structural resilience under complex loading. Reliability-based limit analysis evaluates the lateral load capacity of long piles, incorporating plastic limit analysis and uncertain strain energy bounds, with optimization guided by a reliability index and Gaussian-distributed uncertainties [10, 11]. A steel silo damaged in the 2011 Van earthquake revealed inadequate welding and fusion as primary causes of collapse, emphasizing the importance of strong tank-column connections to mitigate earthquake damage.

Furthermore, the previously mentioned incident in Ferrybridge, UK [1] occurred due to the strong wind (velocity estimated to be 19 m/s). Three out of eight reinforced cooling towers fell in this incident. Therefore, as more and more higher cooling towers are being built, and the wind load also has become one of the controlling factors while designing a cooling tower. One of the first field analyses and measurements was conducted in the 1960s, in the four-tower arrangement of the West Burton power station in the UK [12].In the combined effects of the temperature, gravity loads, seismic loads, soil condition, soil structure interaction, the wind load is the dominating factor [13–16].In the study, [17] the wind pressure distributions on the internal and external surfaces of the cooling towers were observed and analyzed. The tower on which the research was based, was 113.5 m high. 72 points and 36 points were taken as measuring points in both the internal surface and the external surface. More research of this kind was conducted which involved the analysis of pressure distributions along the cooling tower surface [18–20]. Often, the research on the structural and dynamic response of towers due to the wind loads has been overlooked. In 1980, dynamic response analysis was conducted on the three cooling towers including the Weisweiler, Matin Cereek and Schmeehausen cooling towers in the paper [21]. This study discussed the determination of the gust factor along with the result of the dynamic wind load analysis. Furthermore, the wind-induced collapse of the cooling tower was analyzed [22] for the evaluation of realistic safety factors.

Some researchers have developed protective devices and advanced techniques for such structures in this field. Numerical models for fluid-viscous dampers and High Damping Rubber Bearings are compared, with differential models proving more accurate and robust. Emphasis is placed on balancing accuracy and stability to address production variability, ensuring reliable seismic performance in real structures[23, 24]. Various control strategies, including hybrid systems and Bouc-Wen-based semi-active control, are evaluated for mitigating wind and seismic effects in long-span bridges, highlighting improved robustness, adaptability, and reliability in structural control [25, 26]. The IRIS reactor's safety-by-design approach emphasizes seismic risk reduction via base isolation of its NSSS building, with refined fragility analysis and a numerical method using Response Surface and Monte Carlo simulations to assess High Damping Rubber Bearings under nonlinear dynamics [27, 28].

Despite the importance of research into the structural response of cooling towers, which are thin shell structures, there's a lack of numerical research for nuclear power plant. The reason for this lies in the difficulty of the numerical modelling of such megastructures and the computational cost behind them. In most of the cases of numerical modelling, the shell is modelled with the shell elements and the columns are modelled as solid elements that represent the columns as uniform units, which, they are not. Therefore, often the columns and shell were modelled in a more simplified procedure to reduce the computational cost and make the numerical models simpler. In the following papers, [15, 29, 30] the element BEAM 188 which is a Timoshenko Beam type, was used to represent the columns in ANSYS software. Also, SOLID 45 was used in some papers to represent the column of the tower [4]. Advanced finite element formulations for Timoshenko beam models are developed to simulate reinforced concrete under static loading, capturing diffuse and localized failures through embedded discontinuities and cohesive laws. A multi-fiber beam element enables efficient crack and hinge modeling, and comparative analysis of displacement-based formulations identifies an improved method with enhanced accuracy, particularly in axial-bending interactions [31–33]. On the other hand, some researchers included more detailed modelling of their columns including the embedded steel bars in the concrete column

[34, 35]. One of the studies introduces the deep embedment (DE) method to strengthen RC beams with limited web access, achieving up to 23.29% increased shear capacity and showing strong correlation with FE analysis results [36]. Another study demonstrates that concrete resistance can be reliably estimated using young's modulus derived from FE model updating with ambient vibration data, showing good agreement with destructive test results despite higher variance in the latter [37].

There is limited evidence in the literature examining the impact of choosing two different approaches to modelling methods. Numerical models with the columns with embedded steel bars in concrete would produce different results in columns that are modelled in a more simplified way by using the beam element or the solid elements of the traditional FEA software. To address this issue, this study considers two models with two different approaches for column modeling. In one of them, columns were modelled with the more simplified approach and in another the columns were modelled realistically with the steel bars embedded in the concrete. Both models were tested by analyzing their natural frequencies, and by their response to non-linear analysis of seismic and wind loads. The results of this research may help others in their judgement while making numerical models for their desired outcome.

2 Geometrical and material description of the tower

The geometrical requirements of a hyperbolic cooling tower are extremely complicated, so the analysis of such towers is very sophisticated. Therefore, a finite element modeling approach is taken for a simpler way of analysis. The tower analyzed in this research was one of the four towers at Rooppur Nuclear Power Plant. The tower is built upon 100 Λ-shaped RCC columns. The tower's shell has a hyperbolic shape. It's governed by the following equations:

$$R_{bottom} = 119892 + 26511 \times \sqrt{1 + \left(\frac{h - 125}{741962}\right)^2}\ from\ elev. + 10300\ mm\ to 125000\ mm \tag{1}$$

$$R_{top} = 306721 + 78282 \times \sqrt{1 + \left(\frac{h - 125}{669718}\right)^2}\ from\ elev. + 125000\ mm\ to 175000\ mm \tag{2}$$

The tower has a height of 175 m. The shell has an equal thickness of 500 mm throughout the tower. The shell is reinforced with two rows with separate bars in meridian and hoop directions. There are meridional hoop bars of 12 mm diameter and hoop bars of 16 mm in the tower. In both directions, the cover is 40 mm. The geometry of the tower in Table 1 and the geometric representation is shown in Fig. 1.

Table 1 Geometry value of the tower

Parameter name	Value
Height (mm)	175,000
Height from ground to shell (mm)	10,300
Height of Throat (mm)	123,240
Angular spacing of column	7.2°
Diameter at ground (mm)	128,168
Diameter at throat (mm)	76,824
Diameter at top (mm)	81,130

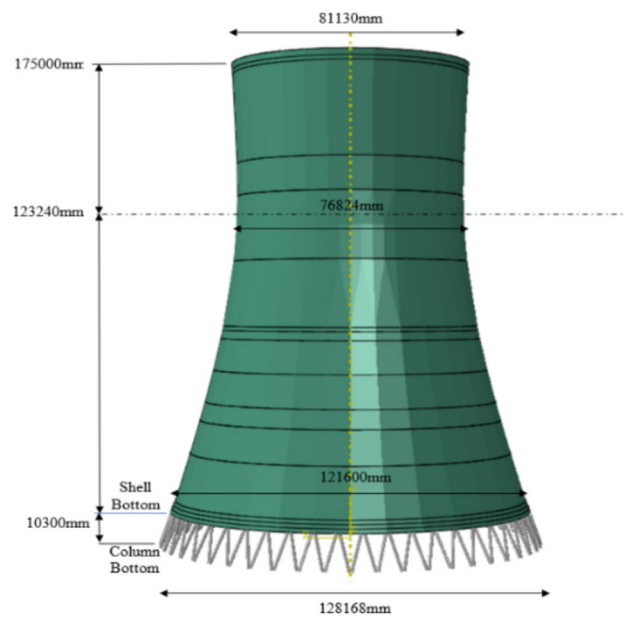

Fig. 1 Geometric representation of tower [38]

Each of the 100 columns creates an angle of 7.2° with the longitudinal axis of the tower. The columns are inclined towards each other at an angle of $\omega = 13.195°$ and at an angle of $\alpha = 17.54°$ towards the inner side of the tower with respect to the meridional axis. The length of the column is 13280 mm. The diameter of the column cross section is 800 mm. The longitudinal bars had a diameter of 32 mm,

Table 2 Details of reinforcement

Reinforcements details	
Items	Dimension (mm)
Hoop bar diameter	16 mm
Meridional bar diameter	12 mm
Composite column longitudinal bar diameter	32 mm
Spiral bar diameter	12 mm

Table 3 Property of steel

Steel property	
Yield strength	570Mpa
Elastic modulus	200Gpa
Poisson's ratio	0.3
Plastic strain	0
Density	7800 kg/m^3

Table 4 Property of B30 concrete

Concrete property			
Material's parameters	B30 [39]	Plasticity parameters	
		Dilation angle	31
Concrete Elasticity		Eccentricity	0.1
E (GPa)	26.6	fb0/fc0	1.16
Poisson's ratio	0.2		
		K	0.67
		Viscosity	0

Table 5 Property of simplified column

Simplified column property	
Mass density (kg/m^3)	$2.5 \times 10^{-9} kg/m^3$
Young's Modulus (MPa)	28,397.44
Poisson's ratio	0.2

and the spiral bars had a diameter of 12 mm. A clear cover of 63 mm is provided for the column.

The reinforcement for shell and column is of grade A240 as per GOSf 578 l-82 and A500 as per GOST R 5254&-2006 as shown in Table 2.

The property of the steel is shown in the Table 3. The concrete for the shell should be of Russian B30 grade as shown Table 4. The concrete damaged plasticity model CDP was followed for the concrete properties. The properties for the simplified column modeled as Timoshenko beams are shown in Table 5.

3 Modelling of the tower

A FEM-based model of the cooling tower was made to analyze the response due to dynamic loads like earthquakes and wind loads. This was achieved by using the commercial finite element software ABAQUS. For both earthquake and wind loading, a total four were created. One of the models for the earthquake loading has embedded reinforcement columns and the other one has simplified column modeled as Timoshenko beam. For wind analysis, two models were created for two types of columns, and the shell was divided into 17 levels.

In Abaqus, the hyperbolic shape was created by using Python scripts following Eqs. (1) and (2). As the tower itself is a thin shell structure, the conventional shell provided by the Abaqus software was used for the modeling of the tower. Conventional shells perfectly represent the thin wall behavior of such large structures. This type of shell is also computationally economical and simpler to analyze (Fig. 2).

The two towers with which the wind analysis was conducted were divided into 17 levels Fig. 3 in accordance with wind loading factors stated in BNBC-2020. The other two towers used in the earthquake analysis were not divided into separate sections or levels. Mesh details of all four towers are shown in Table 6. In the modeling of the shell, the S4R

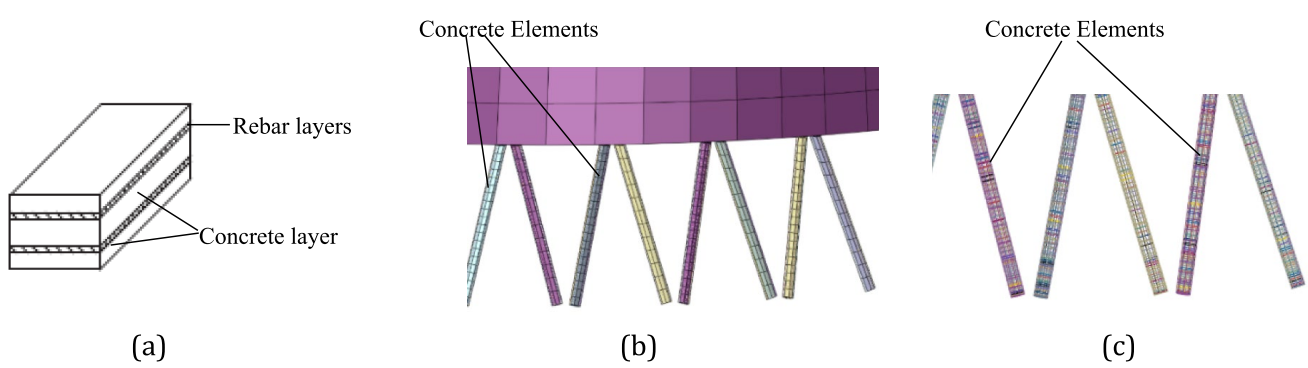

Fig. 2 (**a**) Conventional shell element (**b**) Concrete elements (**c**) Reinforcing steel element

Fig. 3 Seventeen layers for wind load application

shell element was used. S4R is a 4-noded quadrilateral shell element that is allowed to be used in static and dynamic procedures. The meridional and hoop reinforcement bars were used in the conventional shell using the "Rebar Layers" option in Abaqus.

The RC columns were modeled realistically by embedding the longitudinal bars and spiral bars in the concrete column as seen in Figs. 4a and 2. This type of column is not computationally efficient and necessitates a lot of computational power. Here, the concrete column is meshed with C3D8R solid elements. The bars and spiral bars are meshed with T3D2 elements. On the other hand, for the other models, the columns are simply modeled as Timoshenko beam element B31as shown in Fig. 4b**.**

Table 7 Details of Bhuj city earthquake

Bhuj City Earthquake	
Latitude	23.0300°
Longitude	72.6300°
Source	Dept of Earthquake Eng., Indian Inst. of Technology, Roorkee, India
Location of Ground Zero	Bhuj/Kachchh, India 2001–01-26 03:16:40 UTC 239.0 km

3.1 Earthquake loading

Non-linear time-history analysis was done in the models. For the analysis a certain earthquake that occurred in Bhuj/Kachchh, India at 2001-01-26 03:16:40 UTC was considered.

With the collected data as shown in Table 7, a ground acceleration vs. time graph was created in Seismosignal software. The graph obtained from Seismosignal, which denotes the ground acceleration about the X-axis, is shown in Fig. 5.

Table 6 Details of mesh

	Items	Earthquake loading	Wind loading
Conventional shell	Element	S4R	S4R
	No. of Elements	5936	6136
	Mesh Size at top portion	2.41 m × 2.98 m	2.44 m × 3 m
	At throat	2.866 × 2.988	2.36 m × 2.74
	At bottom	3.57 × 2.95	3.67 × 2.86
Concrete column	Element	C3D8R	
	No. of elements	19,200	
Spiral and Longitudinal rebars	Element	T3D2	
	No. of Elements	286,800	

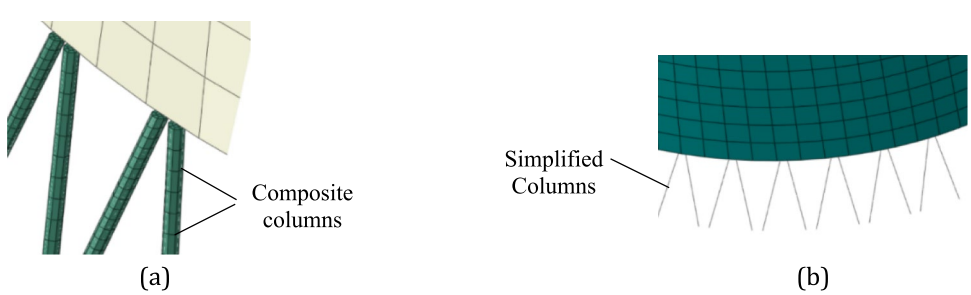

Fig. 4 (**a**) Composite columns (**b**) Simplified columns by "Timoshenko beam element"

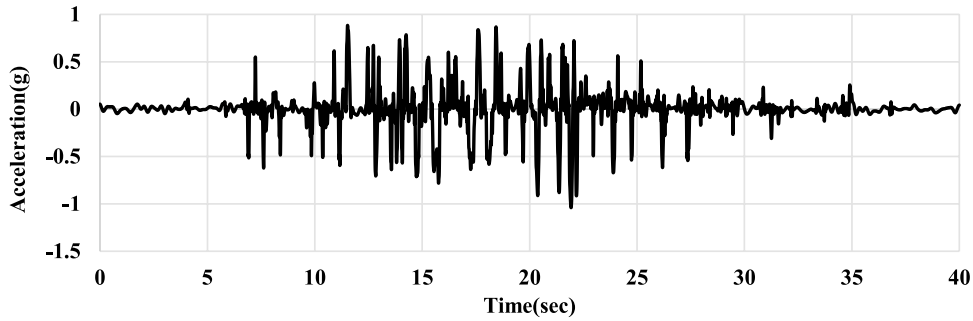

Fig. 5 Ground Acceleration in *X*-axis

3.2 Wind loading

Wind load vertical effect is mostly influenced by the angular distribution. Wind loads created vertically are determined by the soil surface roughness and atmospheric conditions, while wind loads generated angularly are determined by the surface roughness of the structure. In order to analyze how cooling tower constructions, behave when exposed to wind, wind loads are calculated using the ASCE 7–10 or BNBC (2020) standard. According to this code, the following equation represents the cooling tower's effective wind pressure:

$$P(z) = q_z G C_{pe}(\theta) \tag{3}$$

$$q_z = 0.613 K_z K_{zt} K_d I V^2 \tag{4}$$

Empirical evidence has shown that the Fourier cosine series is the preferred choice for defining circumferential pressure patterns on shells, as it effectively reflects this pressure using cosine families. The wind pressure coefficient is determined by employing the Fourier series decomposition:

$$C_p(\theta) = \sum_{0}^{m} a_i \cos(i\theta) \tag{5}$$

θ represents the angle measured from the windward direction, with θ = 0° indicating windward and θ = 180° indicating leeward. The Fourier coefficients are denoted by $C_p(\theta)$. Multiple suggestions have been put forward regarding these coefficients. EN1991-1-4 [41] has equations for determining specific values of the distribution depending on different factors, however it does not furnish the Fourier coefficients required for deriving these equations.

This study adhered to Greiner's proposition [42] as it appears to provide a more precise approximation of the form of the EN1991-1-4 pressure distribution using the given expression:

$$C_p(\theta) = -0.55 + 0.25\cos(\theta) + 1\cos(2\theta) + 0.45\cos(3\theta) - 0.15\cos(4\theta) \tag{6}$$

The details of the wind loading factors are shown in Table 8. The value of K_z was calculated from Table 6.2.11 for various levels by linear interpolation. The value of K_z varying with elevation is shown in Fig. 6.

Table 8 Details of specification of wind pressure

q_z	Wind pressure	Value	BNBC (2020) [40]
G	Gust Factor	0.85	
$C_{pe}(\theta)$	Distribution of angular pressure		
K_z	Velocity pressure effect coefficient		Table 6.2.11
K_{zt}	Topography Factor	1	Table 6.1.10
K_d	Wind directionality coefficient	0.95	Table 6.2.12
I	Importance factor	1.15	Table 6.2.9
v	Velocity	63.1 ms^{-1}	Table 6.2.8

4 Result and discussion

First, a free vibration analysis was conducted to identify the lateral and circumferential modes of the cooling towers. Next, a non-linear time history analysis was performed for earthquake analysis and a wind analysis was carried out to determine the displacement and membrane stress.

4.1 Free vibration analysis

From Table 9, the first ten modes of both models can be observed for the both Embedded Reinforcement Column and the Simplified column. The simplified model indicates slightly larger values for free vibration analysis. As for the difference between the first and tenth modes, Embedded columns show a 38.4367% increase in frequency and the Simplified Columns show a 38.708% change increase in frequency. Therefore, there's a difference of 0.2713% in the increase of frequency. This slight increase in frequency

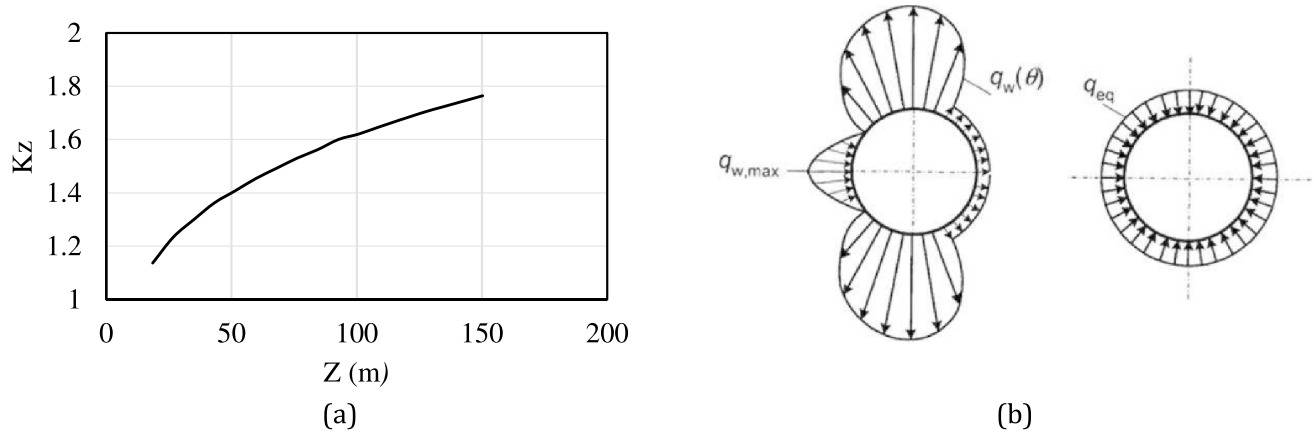

Fig. 6 (**a**) Velocity coefficient with respect to elevation (**b**) Pressure distribution in circular structures [43]

Table 9 Vibrational modes of the tower

Embedded Reinforcement column		Simplified column	
Mode No	Frequency (Hz)	Mode No	Frequency (Hz)
1	0.7628	1	0.8329
2	0.8024	2	0.8746
3	0.8459	3	0.8763
4	0.8485	4	0.8935
5	0.9188	5	0.9706
6	0.9233	6	0.9836
7	0.9334	7	1.0006
8	0.9351	8	1.0016
9	1.0548	9	1.1546
10	1.0560	10	1.1553
Lateral mode			
Mode No	Frequency (Hz)	Mode No	Frequency (Hz)
21	1.4085	21	1.5859
22	1.4118	22	1.5897
42	2.1081	44	2.3858
43	2.1082	45	2.3874

happened because the Embedded Reinforcement Column is less stiff than the Simplified Column.

The frequency associated with the first lateral mode is important for the analysis of earthquake loading [44]. Based on Table 9, the frequency of the first lateral mode of the Embedded Reinforcement Column is 1.4085 H_z and for Simplified Columns, it's 1.5859 H_z as given below in Fig. 7. This indicates a 12.6% increase in frequency. Therefore, if the columns of the tower are modelled as the Embedded Reinforcement Columns which are more realistic, the results will show the cooling tower as less stiff. On the contrary, modelling the columns with Simplified Column provides results with more frequency and thus more stiffness.

The alteration of the period of vibration and corresponding mode shape resulting from the modification of the tower's modelling method can have a substantial impact on the dynamic response of the overall structure.

Table 9 also presents the vibration frequencies for the circumferential modes. The tower experiences meridian deformation and circumferential deformation as shown in Fig. 7, which are caused by circumferential modes. The most significant aspect is the lowest frequency (with the longest period) of these modes, even if it is associated with a large circumferential wave number. The lower frequencies of the circumferential modes than that of the lateral modes prove

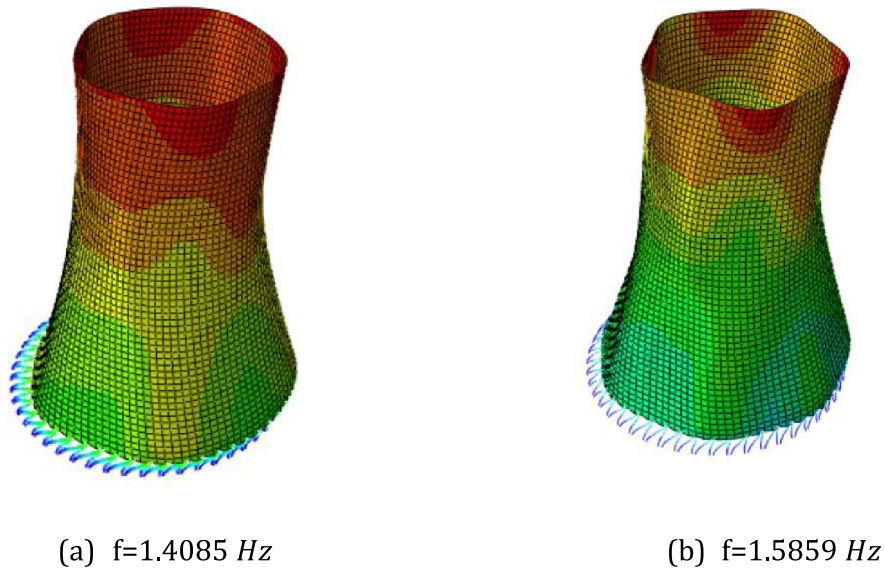

Fig. 7 (**a**) First lateral mode of Embedded Reinforcement Column Model, (**b**) First lateral mode of Simplified Column Model

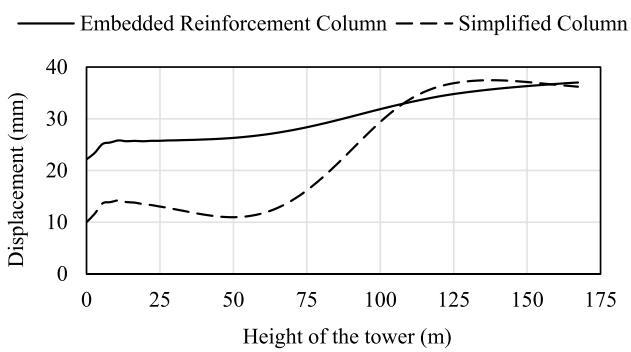

Fig. 8 Maximum radial displacements along the tower height at $\theta = 0$ due to earthquake loading

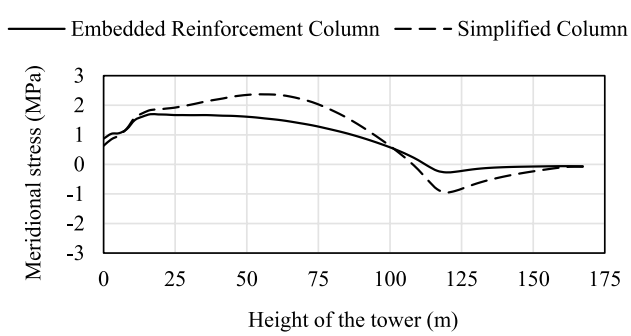

Fig. 9 Meridional stresses along the height of the tower at θ=0° due to earthquake loading

the fact that hyperbolic cooling towers are more prone to wind disturbances than seismic ones. [44].

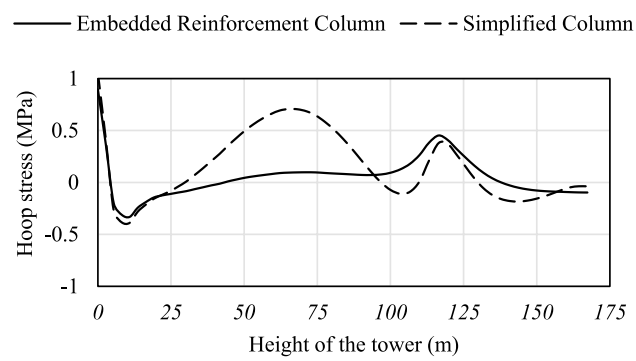

Fig. 10 Hoop stresses along the height of the tower at θ=0° due to earthquake loading

4.2 Earthquake analysis

The results of the earthquake's greatest ground acceleration period were selected in order to analyze the extreme conditions. When subjected to seismic disturbances, columns are typically the weakest component of a hyperbolic cooling tower[4].Due to this fact, there was an abrupt increase in displacement at the bottom of the shell where the columns are connected. For this study, the displacement is measured from the bottom of the shell which is represented by the origin of the X and Y axis in Fig. 8.

The Embedded Reinforcement Column model showed more displacement than the model with Simplified Columns. However, the maximum observed displacement was 38 mm, which remains below the 52 mm displacement reported for a 176 m tower (Λ-75) in previous studies [45]. This was due to the formation of cracks and plastic hinges in the embedded columns. These plastic hinges and cracks made the model flexible therefore the stiffness decreased

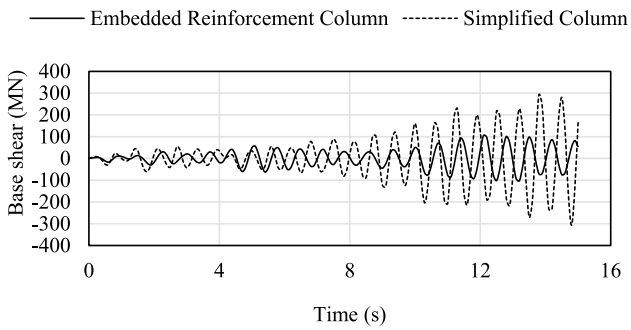

Fig. 11 Base shear force under earthquake excitation

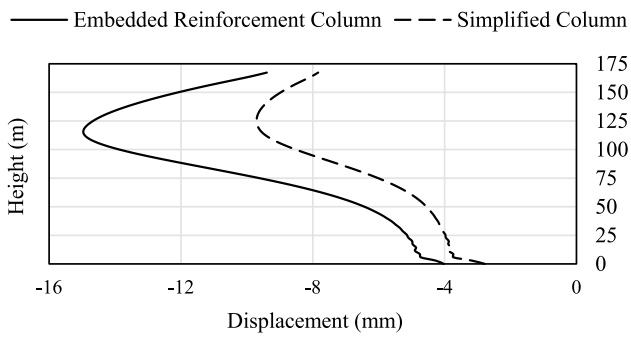

Fig. 12 Maximum radial displacements along the tower height at $\theta = 0°$ due to wind loading

causing large deformations. Thus, the Embedded Column model performed more realistically than the model with the Simplified columns.

In the case of meridional and hoop stresses, both columns performed almost identically Fig. 9. The embedded columns were good at dispersing energy through the creation of cracks, hence the meridional stress for the embedded column model was marginally lower than the simplified column model. The maximum meridional stress recorded was 2.4 MPa, which is notably lower than the value of approximately 5.5 MPa (Λ-75) reported in previous studies [45].

In Fig. 10, the high magnitude of the hoop stress in the column-shell junction occurred due to stress concentrations at the junction points for both models. For the aforementioned causes, the hoop stress of the simplified columns also was greater in magnitude than the model with embedded columns. The maximum hoop stress was 1 MPa, which is significantly lower than the 5 MPa observed in my earlier studies [45].

Because of the inelastic behaviour of the embedded reinforcement columns due to cracks and plastic hinges, the base shear was significantly lower than the model with a simplified model as shown in Fig. 11. In the temporal perspective, the base shear decreased at the end of the selected spectrum as the plastic hinges already formed.

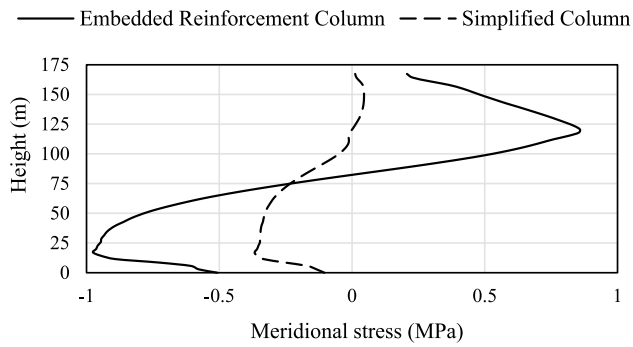

Fig. 13 Meridional stresses along the height of the tower at θ=0° due to wind loading

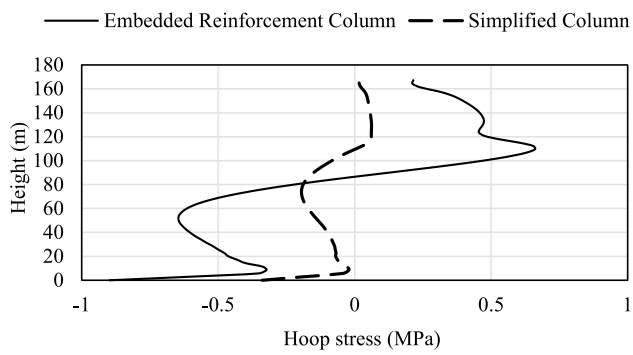

Fig. 14 Hoop stresses along the height of the tower at θ=0° due to wind loading

4.3 Wind analysis

The displacement that occurred due to wind loading in the $\theta = 0°$ direction is shown in Fig. 12. Due to the flexibility of the embedded reinforcement columns, the displacement is higher than the model with the simplified column. For both of the models, the displacement particularly increased in the throat section where the circumferential area decreased thus increasing the effect of the wind force. It was also observed that the model embedded columns provided a maximum displacement of 15 mm which is significantly lower than the 30 mm displacement reported in previous studies for a tower of 125 m height [46].

Meridional and hoop stresses were also determined in the windward direction at the angle $\theta = 0°$. As stated in previous sections, instead of using Equivalent Static Wind Loading (ESWL), Greiner's coefficients were used to project the wind loading onto the tower as shown in Fig. 6b**.** As a result, in this direction, the meridional stress for both models are higher than the hoop stress as seen in Figs. 13, 14. The maximum meridional stress recorded was 0.9 MPa, which is lower than the 2.5 MPa observed in my earlier studies [46].

Also, the lower half of the tower faced tension and the top half faced compression. Particularly, the columns with embedded reinforcements encountered more stress including both the meridional and hoop stress than the tower with simplified columns. Therefore, it can be concluded that the model with embedded reinforcement has a tendency to produce more stresses when faced with wind excitations.

5 Conclusion

In this study, a finite element model of a hyperbolic cooling tower was developed and observed the effect of different column modelling procedures by performing free vibration analysis, seismic analysis, and wind load analysis. The results of the following analyses in terms of displacement, stress, etc., were compared between the two models. From the analyses, the following conclusions can be drawn:

1. First, the model was tested using the Free Vibration Analysis, also it was found that for the Embedded Reinforcement model the value of natural frequency was less in magnitude than the model with simplified columns. This confirms the fact that the Embedded Reinforcement Column model was more flexible than the model with simplified columns.
2. During the seismic excitation, the Embedded Reinforcement Column model performed more realistically as cracks and plastic hinges were observed in the columns thus resulting in increased values of displacements than the model with simplified columns. The values of base shear and stresses were also found to be lower for the Embedded Reinforcement Column due to the creation of plastic hinges and cracks which help to dissipate the energy more effectively.
3. As seen in the seismic analysis, the displacements of the Embedded Reinforcement Column model were greater than the columns with simplified columns due to prior reasons. But, in terms of stress, the model with simplified columns showed more stress than the other model.
4. The Embedded Reinforcement Column required significantly larger computational power than that of the Simplified Model.

In terms of dynamic analysis of this type of structure, this study suggests considering the results stated above while selecting a suitable modelling approach. Additionally, further in-depth research on wind loading, particularly using Greiner's equation, as well as the inclusion of soil-structure interaction, is recommended for future studies. Moreover, the exclusion of thermal and operational loads is acknowledged as a limitation, offering another avenue for further investigation.

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Replication of Results The numerical simulations in this study were conducted using Abaqus. Due to confidentiality restrictions, the cooling tower data cannot be shared. However, to facilitate replication, the Abaqus model (.cae file) and the Bhuj City Earthquake Excel file used in the analysis have been provided.

Author contribution This study was supervised by Dr. Md Robiul Awall. The conceptualization, finite element modelling, data analysis, preparation of manuscript was done by Intekhab Uddin Ahmad Chy & Md Asadullahil Galib Fardin. The collection of data and preprocessing was done by Anik Biswas.

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Data Availability The data used in this study are confidential, they are available upon the reviewer's request.

Declarations

Competing Interest The authors declare no competing interests.

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